

Practice Problems Sheet I

(Finite Automata)

Note: Figures are in separate files.

Q1 Consider a neuron net shown in figure 1. All wires carry logical values 0 or 1. The triangle shaped gates (neurons) have two types of inputs (synapses), excitatory shown by hollow circles, and inhibitory shown by filled circles. The output of this gate is 1 iff the number of excitatory inputs with value 1 exceeds the number of inhibitory inputs with value 1 by at least the number written inside the triangle. Consider $y_1y_2y_3$ as state variables with initial values 0, for all of them.

Model this net as a DFA. Input to the DFA is a finite 0-1 stream as it arrives on the wire marked 'In'. The input string is accepted iff output on the wire 'Out' is 1. Assume that there is sufficient time for effects of an input bit to propagate and output of the net to stabilize, before the next input bit arrives.

Q2 For each of the following languages over alphabet $\{0, 1\}$, design a DFA to recognize it.

1. Set of all strings which do not have a substring 1011.
2. $L = \{x \in \{0, 1\}^* \mid \text{Every block of five consecutive bits contains at least two 0's}\}$.
3. Set of all strings which contain an occurrence of *exactly three* (not more) consecutive 0's.
4. All strings in which every 1 is immediately followed by two 0s.

Q3 Describe the language accepted by each of the automaton (a)-(c) in figure 2, in simple English.

Q4 In class, we designed an automaton to accept all 0-1 strings which are binary representation of numbers that leave remainder 1 when divided by 5. In that problem, we

assumed that the input to the automaton is provided in MSB to LSB order. That is, the MSB is input first and LSB is input last. Consider the same problem where input is presented in LSB to MSB order. Design a FA which can recognize language of this version of the problem?

Q5 Let N be a NFA with ϵ transitions. In class, we described a direct construction of an equivalent DFA D , with number of states that is (at most) exponential in the number of states of N . Show how to construct an equivalent NFA N' (without ϵ -transitions) with the same set of states as that of N . Justify the correctness of your construction.

Q6 Design a NFA to find an occurrence of any of the words *lab*, *cab*, *tab*. Let the input alphabet be $\{a, b, c, d, l\}$. Convert your NFA to DFA using subset construction.

Q7 Show that the class of languages recognized by DFAs are closed under the following operations on languages.

1. $Max(L) = \{x \mid x \in L, \text{ no proper extension of } x \text{ is in } L\}$
(w is an extension of x iff $\exists y[w = xy]$. An extension w of x is *proper* if $w \neq x$)
2.
$$Shuffle(L, M) = \{x_1y_1x_2y_2 \dots x_ky_k \mid x_i, y_i \in \Sigma^* \text{ for } 1 \leq i \leq k \text{ and } x_1x_2 \dots x_k \in L, y_1y_2 \dots y_k \in M\}$$
3. $Cycle(L) = \{xy \mid yx \in L, x, y \in \Sigma^*\}$
4. $Half(L) = \{x \mid \exists y \in \Sigma^* [|y| = |x| \wedge xy \in L]\}$

Q8 Let L, M be any languages. For each of the following, prove or disprove the proposed identities.

1. $(L + M)^*M = (L^*M)^*$
2. $(L + M)^* = (L^*M^*)^*$
3. $(LM + L)^*L = L(ML + L)^*$
4. $M(LM + M)^*L = LL^*M(LL^*M)^*$

End